# POWER BI REPORT

## WEEK 7 & 8

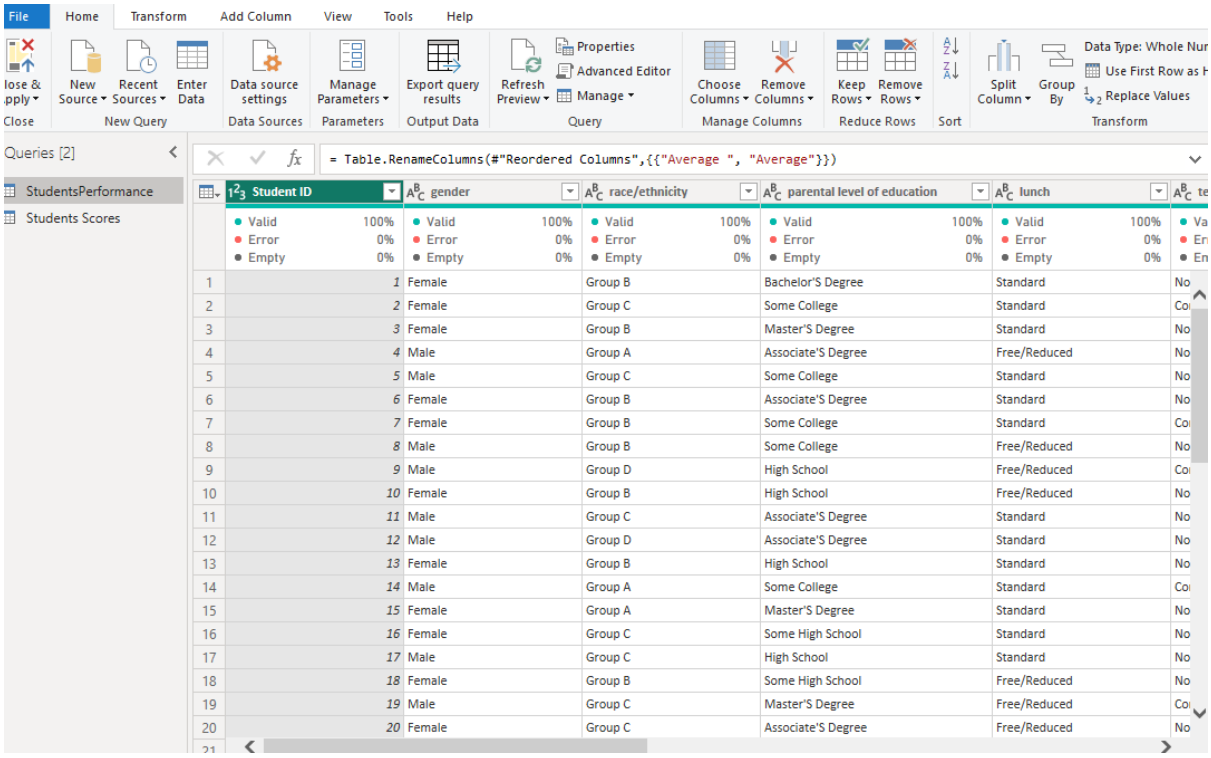
Student performance- Data cleaning & Transformation

**1. Problem Statement**

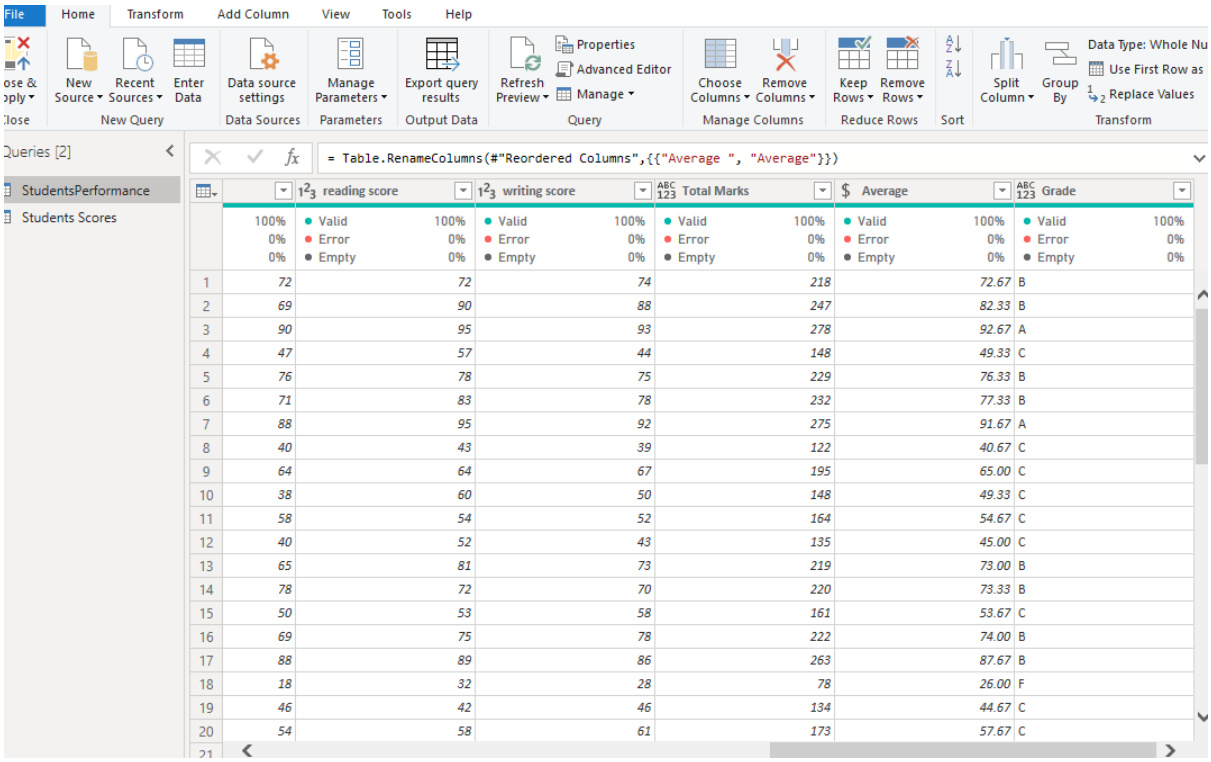
Educational institutions need to understand students' academic performance and identify the factors that may be associated with better or lower results. Raw student examination data can be difficult to analyze because it contains multiple subjects and categorical attributes. This project focuses on cleaning, transforming, analyzing, and visualizing student performance data using Power BI to generate meaningful academic insights.

**2. Objectives**

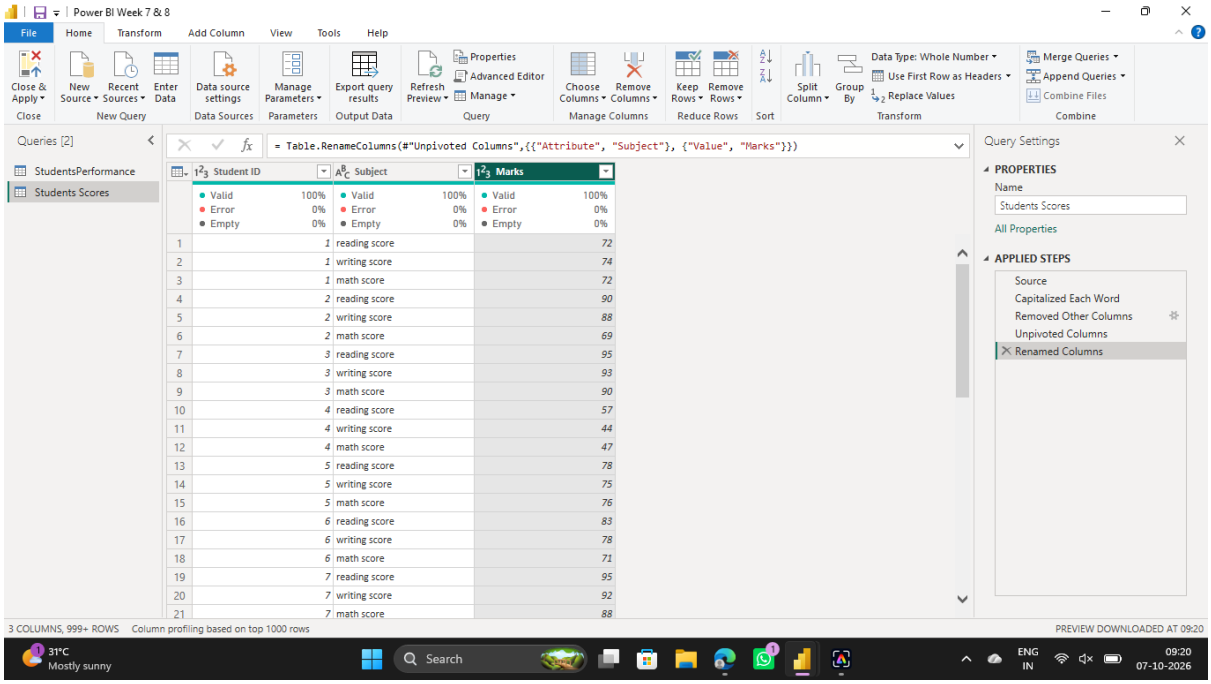
- To import and clean the Student Performance dataset using Power Query.
- To handle missing values, duplicates, data types, and inconsistent categories.
- To calculate total and average student scores.
- To classify students based on performance and Pass/Fail status.
- To analyze performance across Math, Reading, and Writing subjects.
- To study the relationship between academic performance and factors such as parental education, lunch type, gender, and test preparation.
- To create an interactive Power BI dashboard with KPIs, charts, slicers, and drill-through analysis.
- To identify meaningful, evidence-based insights from the dataset.

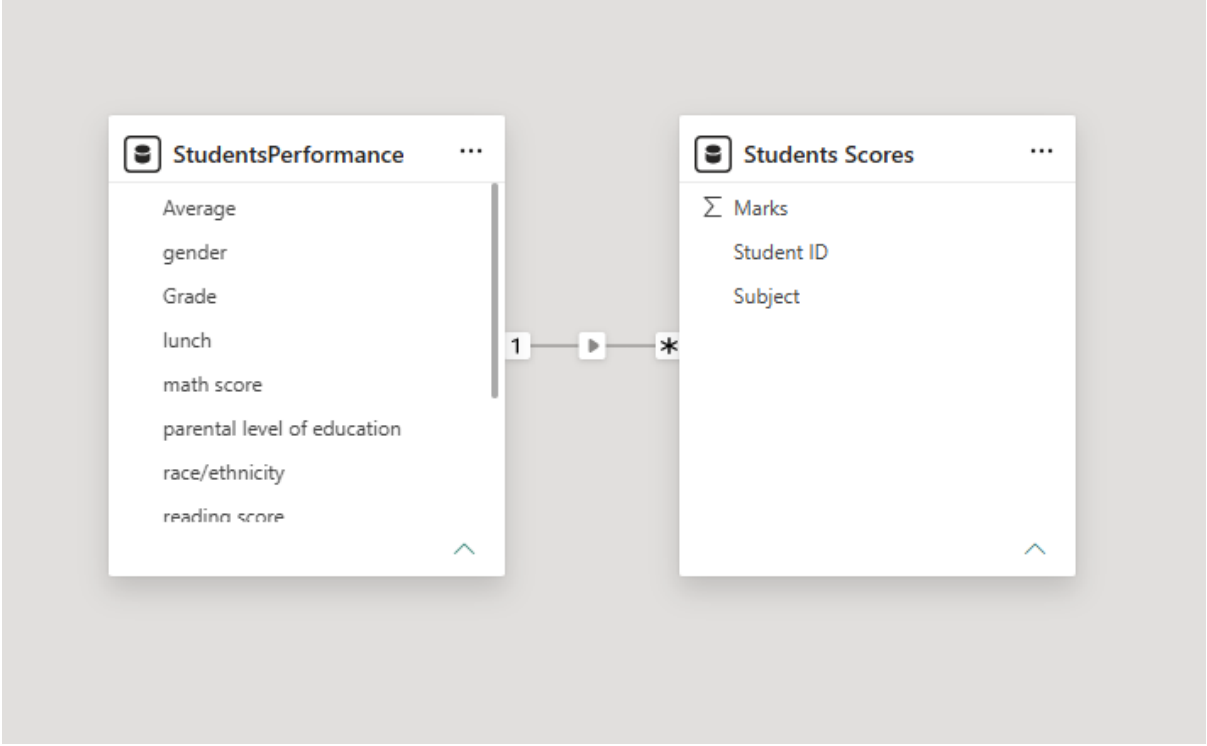


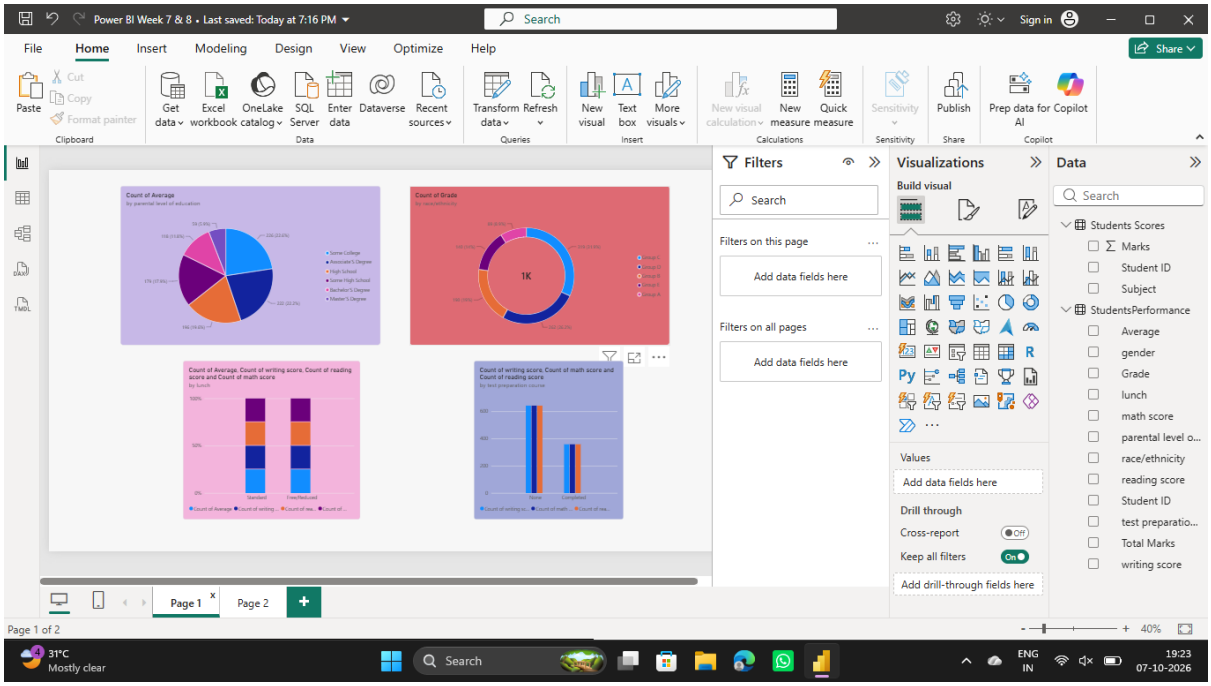

| | Student ID | gender | race/ethnicity | parental level of education | lunch | te |
|---|---|---|---|---|---|---|
| | Valid 100% Error 0% Empty 0% | Valid 100% Error 0% Empty 0% | Valid 100% Error 0% Empty 0% | Valid 100% Error 0% Empty 0% | Valid 100% Error 0% Empty 0% | Va Er En |
| 1 | 1 | Female | Group B | Bachelor'S Degree | Standard | No |
| 2 | 2 | Female | Group C | Some College | Standard | Co |
| 3 | 3 | Female | Group B | Master'S Degree | Standard | No |
| 4 | 4 | Male | Group A | Associate'S Degree | Free/Reduced | No |
| 5 | 5 | Male | Group C | Some College | Standard | No |
| 6 | 6 | Female | Group B | Associate'S Degree | Standard | No |
| 7 | 7 | Female | Group B | Some College | Standard | Co |
| 8 | 8 | Male | Group B | Some College | Free/Reduced | No |
| 9 | 9 | Male | Group D | High School | Free/Reduced | Co |
| 10 | 10 | Female | Group B | High School | Free/Reduced | No |
| 11 | 11 | Male | Group C | Associate'S Degree | Standard | No |
| 12 | 12 | Male | Group D | Associate'S Degree | Standard | No |
| 13 | 13 | Female | Group B | High School | Standard | No |
| 14 | 14 | Male | Group A | Some College | Standard | Co |
| 15 | 15 | Female | Group A | Master'S Degree | Standard | No |
| 16 | 16 | Female | Group C | Some High School | Standard | No |
| 17 | 17 | Male | Group C | High School | Standard | No |
| 18 | 18 | Female | Group B | Some High School | Free/Reduced | No |
| 19 | 19 | Male | Group C | Master'S Degree | Free/Reduced | Co |
| 20 | 20 | Female | Group C | Associate'S Degree | Free/Reduced | No |

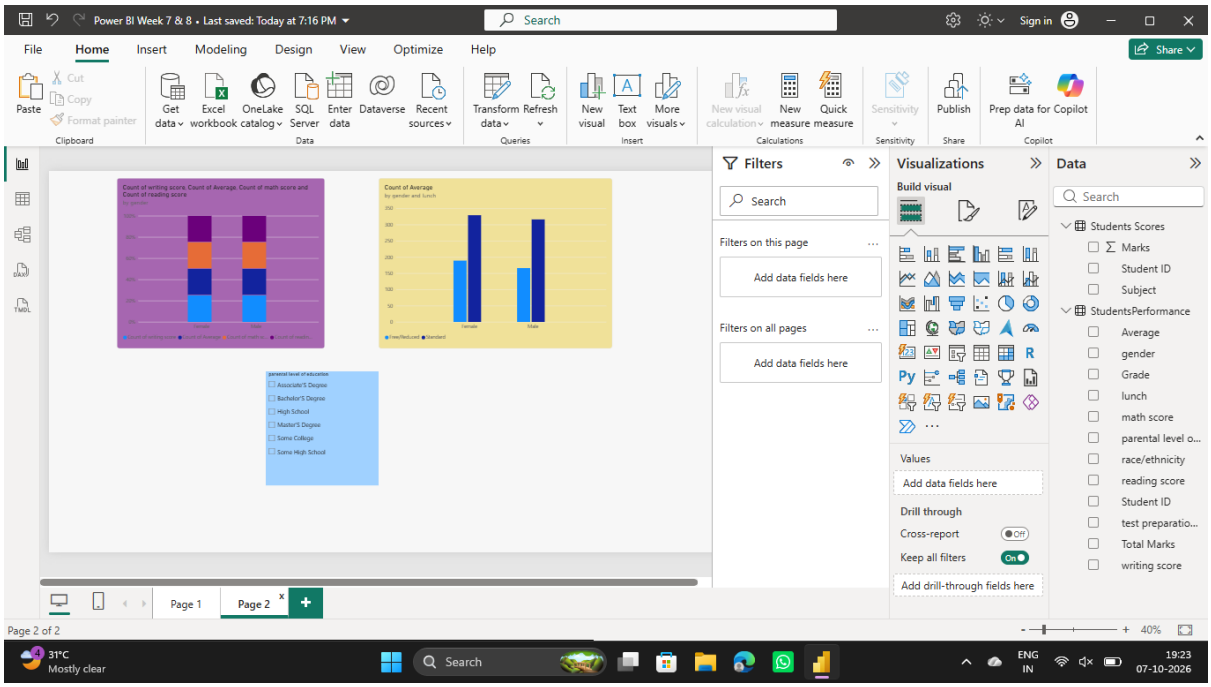



| | reading score | writing score | Total Marks | Average | Grade |
|---|---|---|---|---|---|
| | Valid 100% Error 0% Empty 0% | Valid 100% Error 0% Empty 0% | Valid 100% Error 0% Empty 0% | Valid 100% Error 0% Empty 0% | Valid 100% Error 0% Empty 0% |
| 1 | 72 | 74 | 218 | 72.67 | B |
| 2 | 90 | 88 | 247 | 82.33 | B |
| 3 | 95 | 93 | 278 | 92.67 | A |
| 4 | 57 | 44 | 148 | 49.33 | C |
| 5 | 78 | 75 | 229 | 76.33 | B |
| 6 | 83 | 78 | 232 | 77.33 | B |
| 7 | 95 | 92 | 275 | 91.67 | A |
| 8 | 43 | 39 | 122 | 40.67 | C |
| 9 | 64 | 67 | 195 | 65.00 | C |
| 10 | 60 | 50 | 148 | 49.33 | C |
| 11 | 54 | 52 | 164 | 54.67 | C |
| 12 | 52 | 43 | 135 | 45.00 | C |
| 13 | 81 | 73 | 219 | 73.00 | B |
| 14 | 72 | 70 | 220 | 73.33 | B |
| 15 | 53 | 58 | 161 | 53.67 | C |
| 16 | 75 | 78 | 222 | 74.00 | B |
| 17 | 89 | 86 | 263 | 87.67 | B |
| 18 | 32 | 28 | 78 | 26.00 | F |
| 19 | 42 | 46 | 134 | 44.67 | C |
| 20 | 58 | 61 | 173 | 57.67 | C |


Power BI Week 7 & 8
File
Home
Transform
Add Column
View
Tools
Help
Queries [2]
StudentsPerformance
Students Scores
= Table.RenameColumns(#"Unpivoted Columns",{{"Attribute", "Subject"}, {"Value", "Marks"}})
Student ID
Subject
Marks
Query Settings
PROPERTIES
Name
Students Scores
All Properties
APPLIED STEPS
Source
Capitalized Each Word
Removed Other Columns
Unpivoted Columns
Renamed Columns
3 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 09:20
31°C
Mostly sunny
Search
ENG
IN
09:20
07-10-2026


StudentsPerformance
Average
gender
Grade
lunch
math score
parental level of education
race/ethnicity
reading score
1
*
Students Scores
Marks
Student ID
Subject


Power BI Week 7 & 8 • Last saved: Today at 7:16 PM
Search
Sign in
File
Home
Insert
Modeling
Design
View
Optimize
Help
Share
Paste
Cut
Copy
Format painter
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
New visual
Text box
More visuals
New visual calculation
New measure
Quick measure
Sensitivity
Publish
Prep data for AI
Copilot
Clipboard
Data
Queries
Insert
Calculations
Share
Copilot
Filters
Search
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Students Scores
Marks
Student ID
Subject
StudentsPerformance
Average
gender
Grade
lunch
math score
parental level o...
race/ethnicity
reading score
Student ID
test preparatio...
Total Marks
writing score
Count of Average
Count of Grade
1K
Page 1
Page 2
Page 1 of 2
40%
31°C
Mostly clear
Search
ENG
IN
19:23
07-10-2026


Power BI Week 7 & 8 • Last saved: Today at 7:16 PM
Search
Sign in
File
Home
Insert
Modeling
Design
View
Optimize
Help
Share
Paste
Cut
Copy
Format painter
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
New visual
Text box
More visuals
New visual calculation
New measure
Quick measure
Sensitivity
Publish
Prep data for AI
Copilot
Clipboard
Data
Queries
Insert
Calculations
Share
Copilot
Filters
Search
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Students Scores
Marks
Student ID
Subject
StudentsPerformance
Average
gender
Grade
lunch
math score
parental level o...
race/ethnicity
reading score
Student ID
test preparatio...
Total Marks
writing score
Count of Average
Page 1
Page 2
Page 2 of 2
40%
31°C
Mostly clear
Search
ENG
IN
19:23
07-10-2026

## Conclusion:

The Student Performance Power BI project successfully transformed raw examination data into an interactive and informative dashboard. Data cleaning, transformation, DAX measures, and visualization techniques helped present student performance clearly. The dashboard makes it easier to compare subjects, explore different student groups, and identify patterns in academic performance. Overall, the project demonstrates how Power BI can be used to convert educational data into meaningful insights for analysis and decision-making.